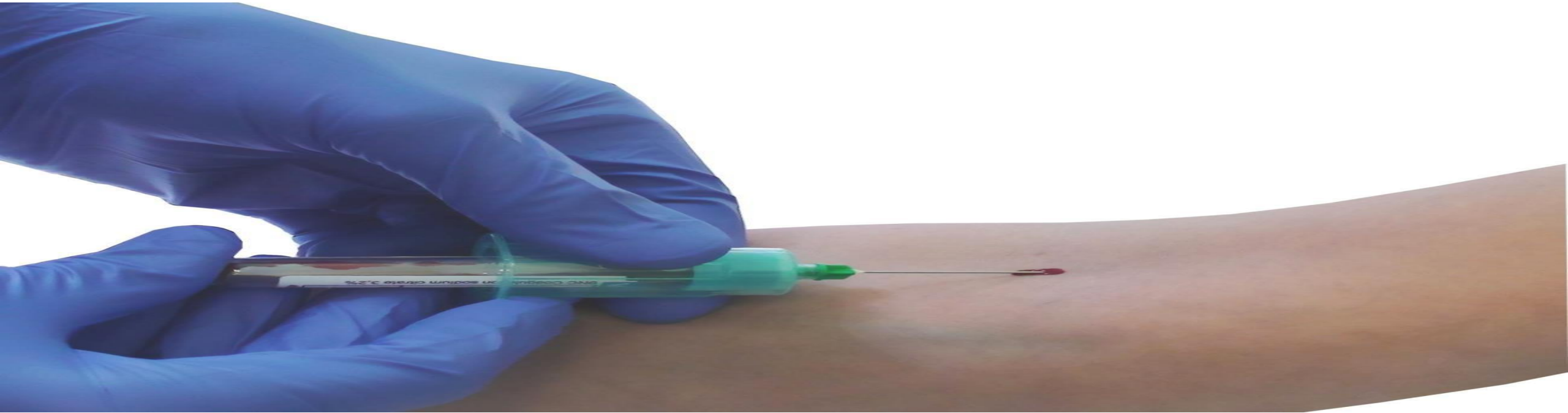


Phlebotomy (Venipuncture)

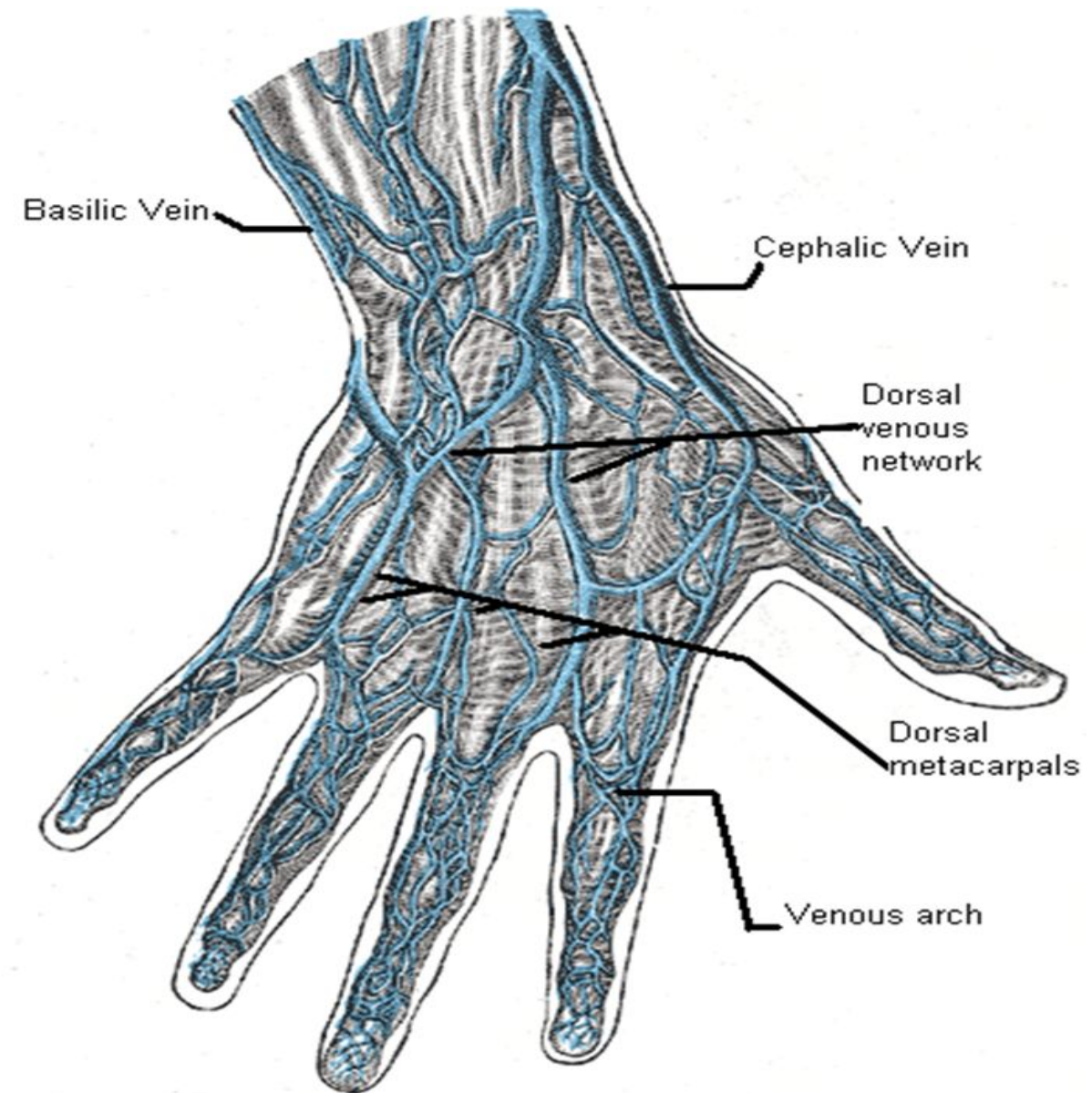
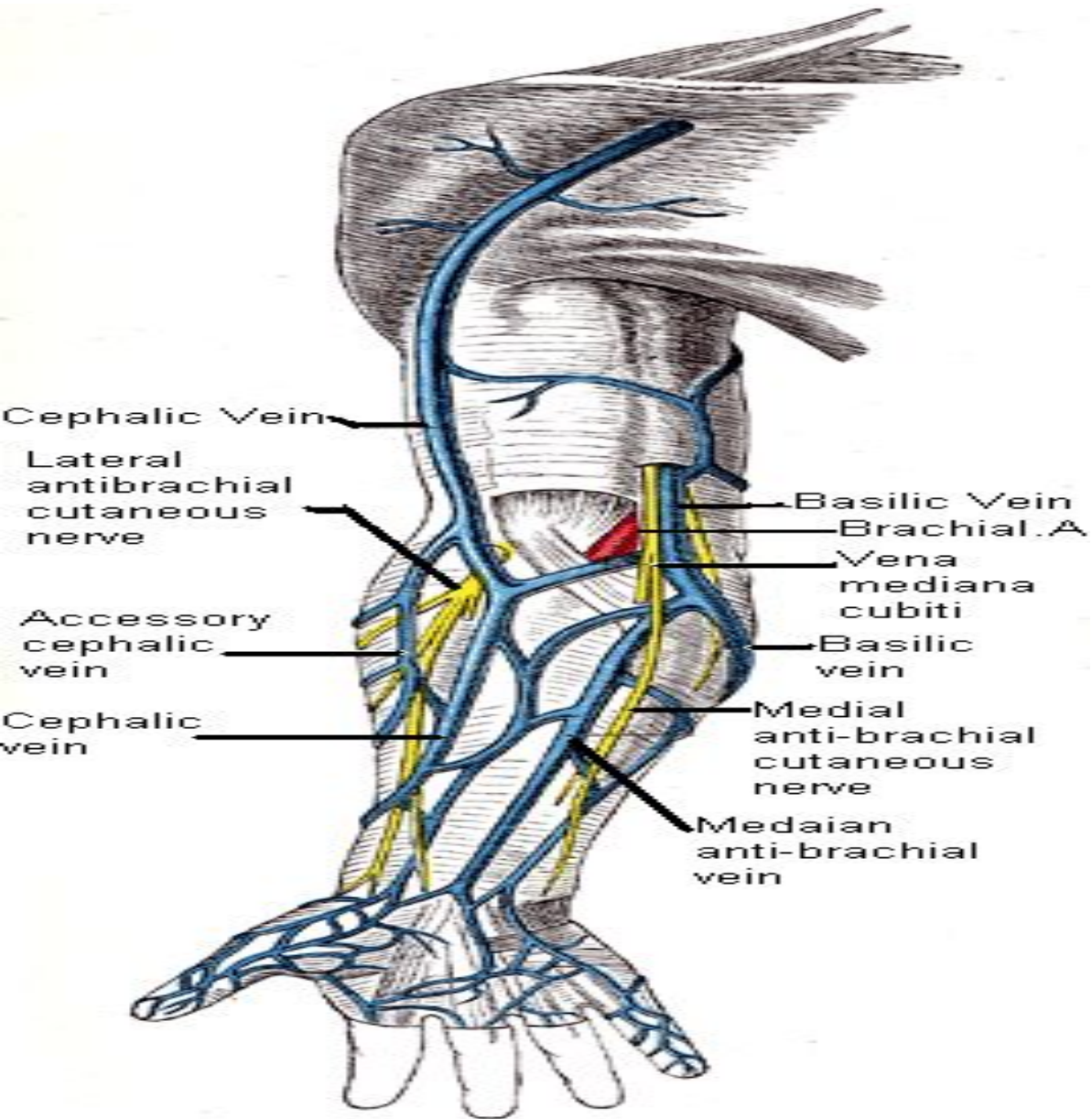


DR ABDUL SAMAD KHAN
SENIOR LECTURER
DIKIOS-DUHS

Learning Outcomes

- Gain an understanding of relevant anatomy
- Identify factors influencing vein selection
- Perform appropriate checks prior to procedure and correct documentation post procedure
- Relate ANTT to performing phlebotomy
- Undergo a simulated assessment using NPSA 'Obtaining a sample for blood transfusion' assessment tool

Anatomy and Physiology



Indications for IV placement or Venipuncture

- 1- Blood sampling , Venous
- 2-Administration of the following fluids
 - Fluid solutions
 - Intravenous medication
 - Blood Products
 - IV Nutrition
- 3-Cardiac Arrest
- 4-Major Trauma
- 5-Cardio Pulmonary resuscitation

Contra Indications

- ❑ Sites Close to infection
- ❑ Veins of fractured limbs
- ❑ Where there is a AV fistula is present
- ❑ Edema
- ❑ Affected site of CVA
- ❑ Site of Mastectomy
- ❑ Extra care should be taken on patients with bleeding, clotting disorders and on Warfarin

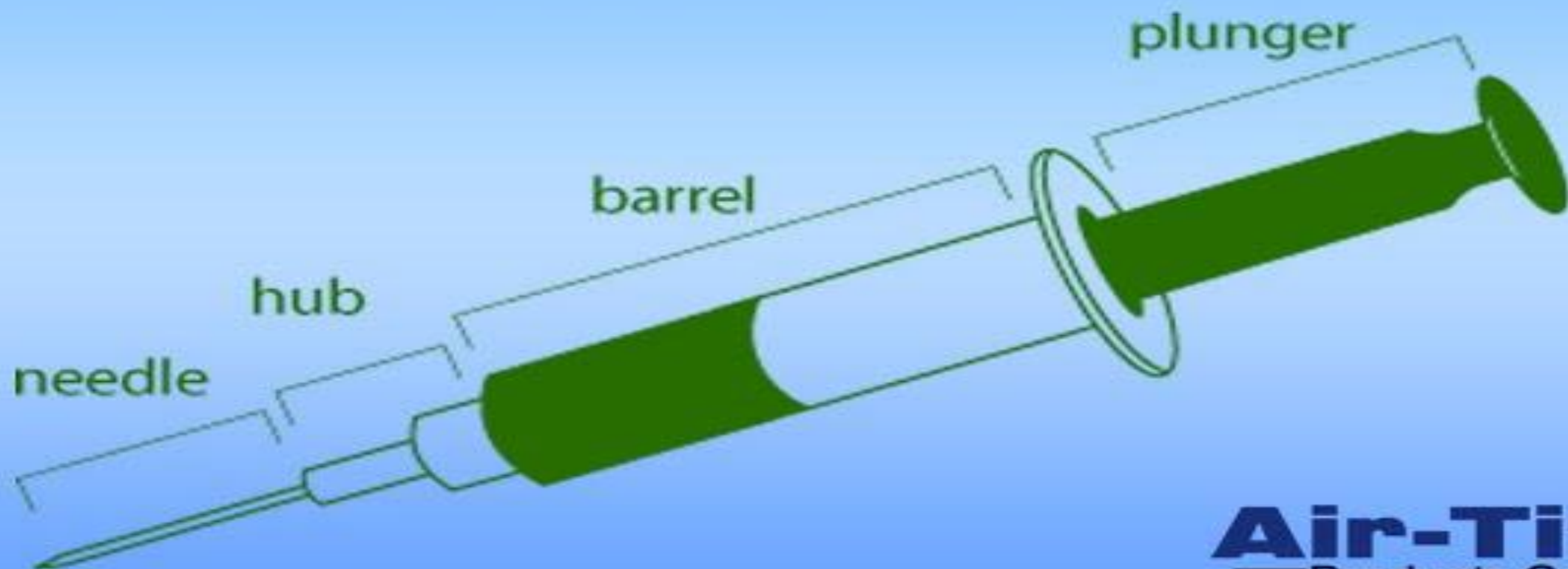
Factors Affecting Vein Selection

- ❑ Previous use and condition of the veins
- ❑ Clinical presentation of the patient e.g. dehydration, shock
- ❑ Patient preference/co-operation/prior experience
- ❑ Medication/on going therapy e.g. I.V infusion, warfarin, steroids
- ❑ Obesity/malnutrition
- ❑ Skin condition

Sites to Avoid

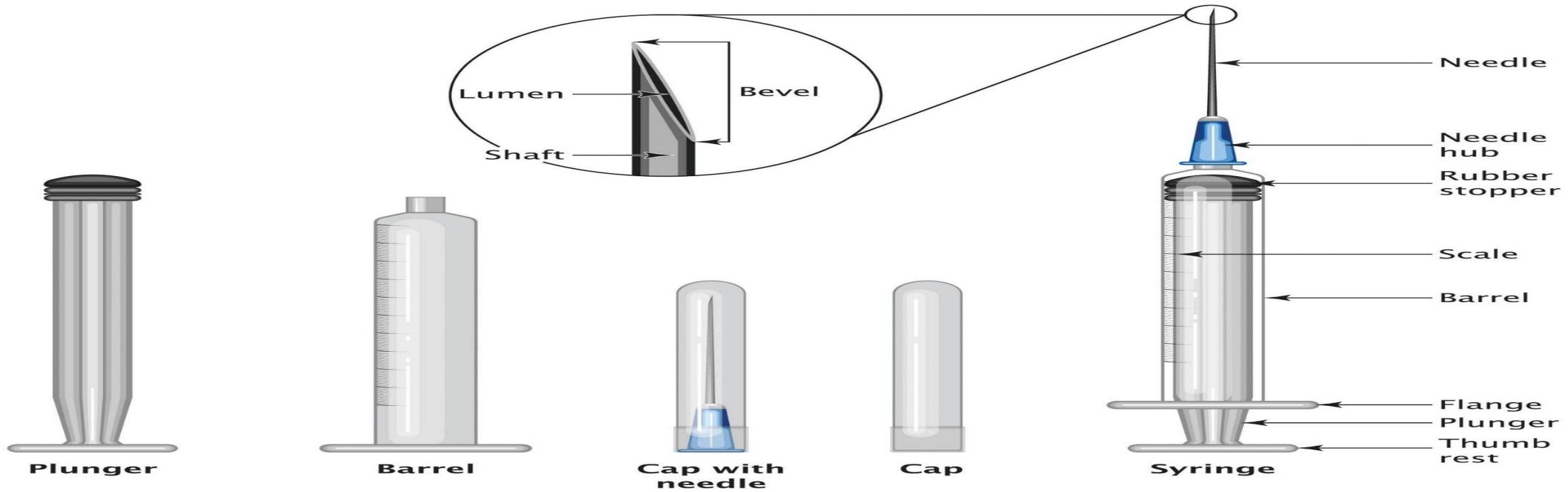
- Sites above I.V. cannula
- Sites where an intravenous infusion is in progress
- The arm on the side of a mastectomy
- Edematous areas
- Areas with significant scar tissue
- Injury:
 - Fractures Paralysis
- Arms with fistulas or vascular grafts

Syringe & Needle Basics



Air-Tite
Products Co., Inc.

Anatomy of a Syringe



SIZE OF THE NEEDLE

1. 16-Gauge (Blood Donor Collection)
2. 21- Gauge (Routine Venipuncture)
3. 23-Gauge (Patient with small veins, butterfly)

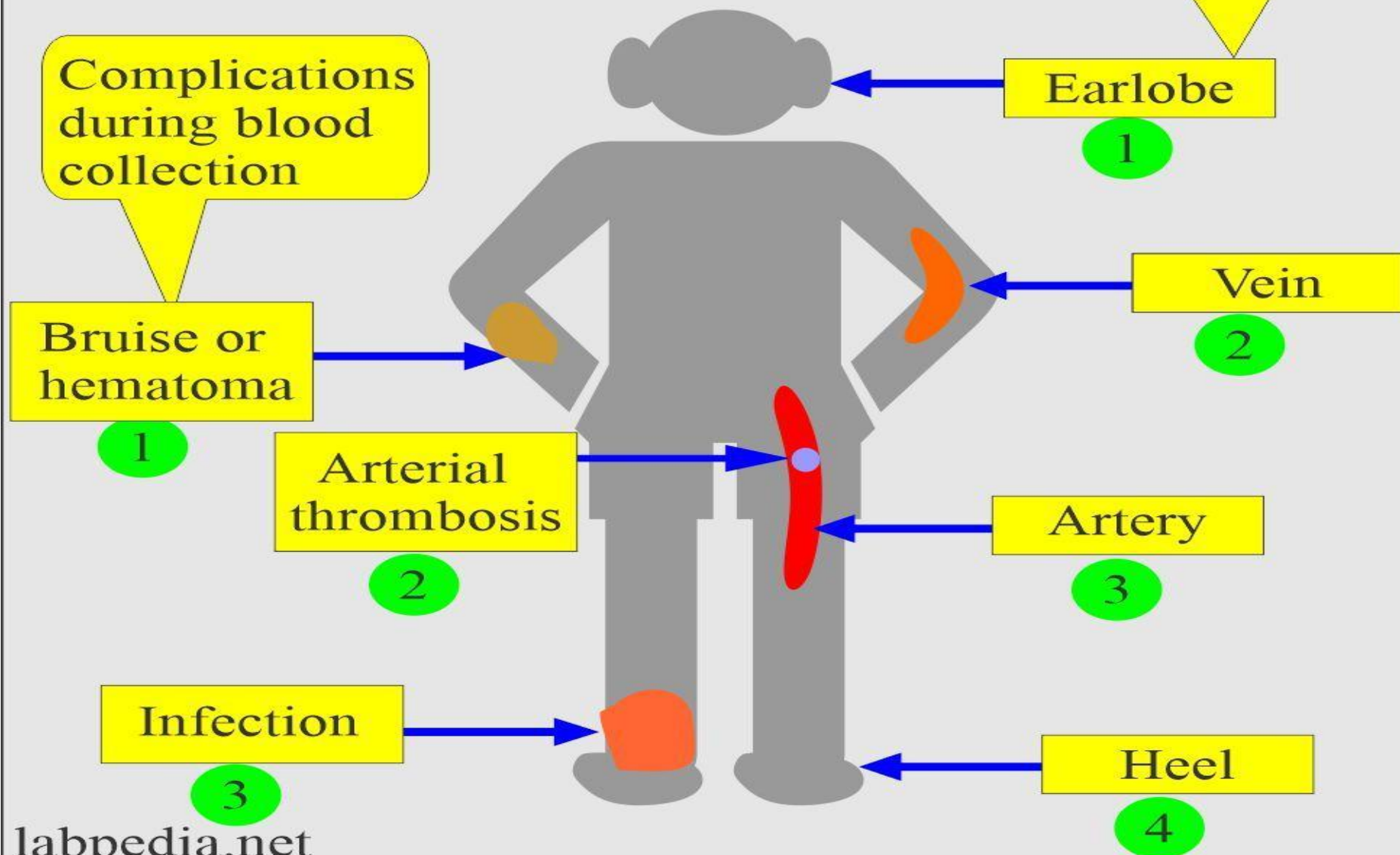


VACUTAINER TUBES

Color	Anticoagulant	Uses
	No anticoagulant	Serological examination in biochemistry
	Sodium Fluoride	Glucose estimation
	EDTA @VijayPatho	Hematological examination like complete hemogram, ESR
	3.2% sodium citrate	Coagulation studies like PT, APTT
	Heparin	Bone marrow studies
	Citrate	Blood culture
	(K2)EDTA	Blood Bank tests. Blood typing. ABO grouping etc

Blood collection sites and complications

Various Sites of blood sample collection

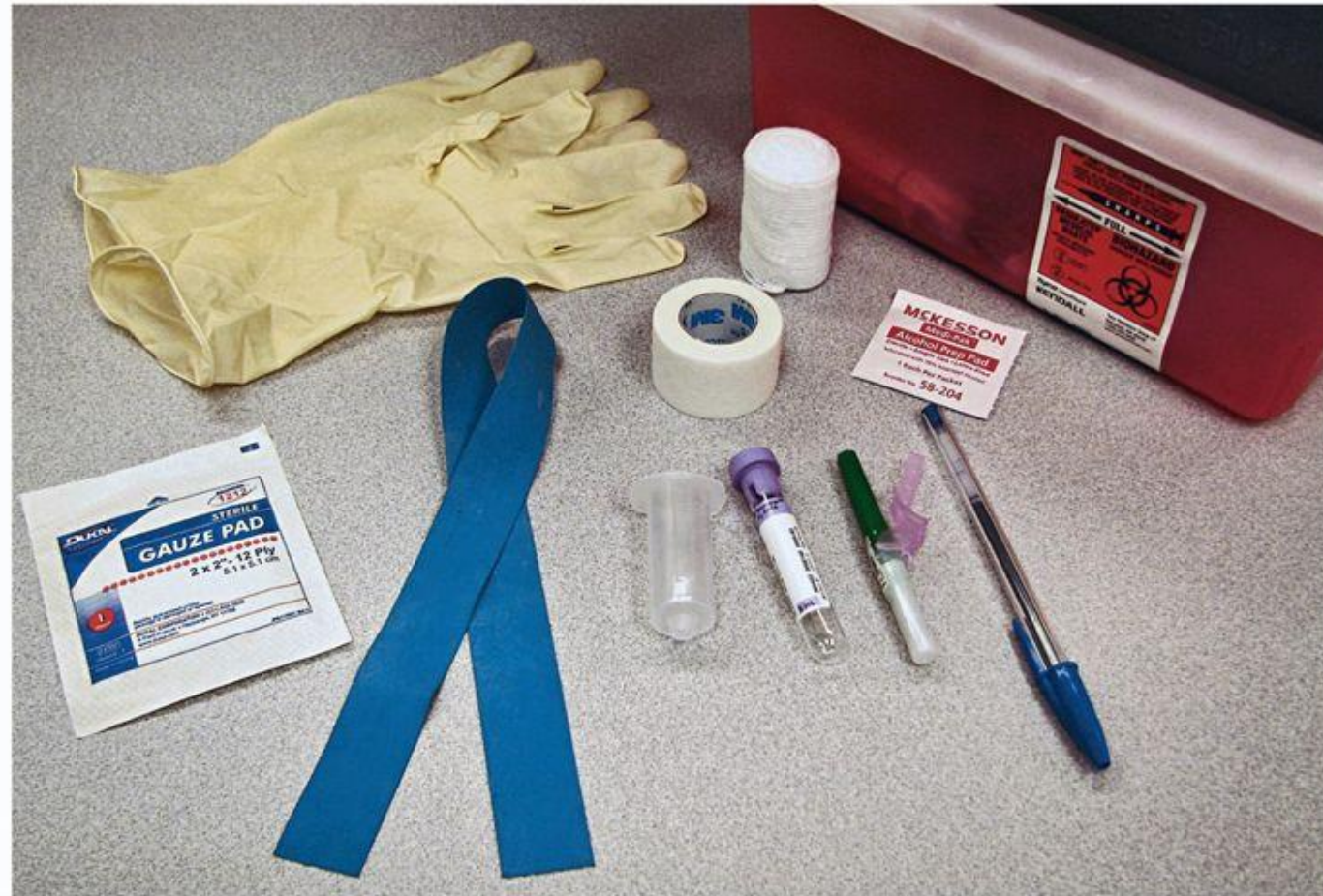


ONLY Sample ONE patient at a time

- ☐ Ensure request has been generated
- ☐ Check patient records
- ☐ Confirm correct patient
- ☐ Vein Selection
- ☐ Preparation of vein
- ☐ Capacity/Unconscious
- ☐ Gain informed consent
- ☐ Assess site suitability
- ☐ Urgent samples priority

Assembling Venipuncture Equipment

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Information that must be on a patient's ID band

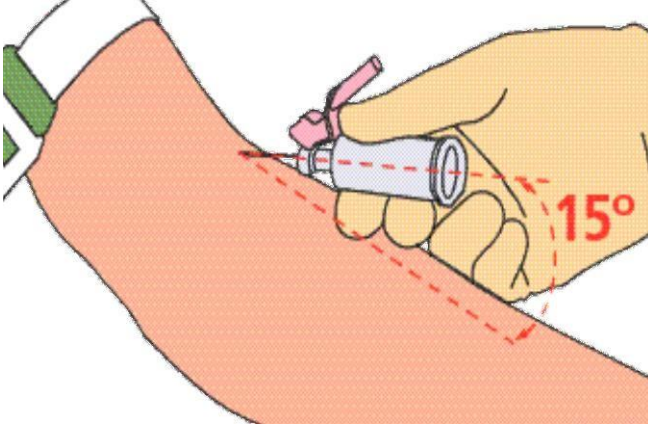
- Patient's Name
- Patient's ID Number
- Date of Birth
- Requesting Physician
- Tests requested
- Date and Time

Where to look for ID Band on the patient

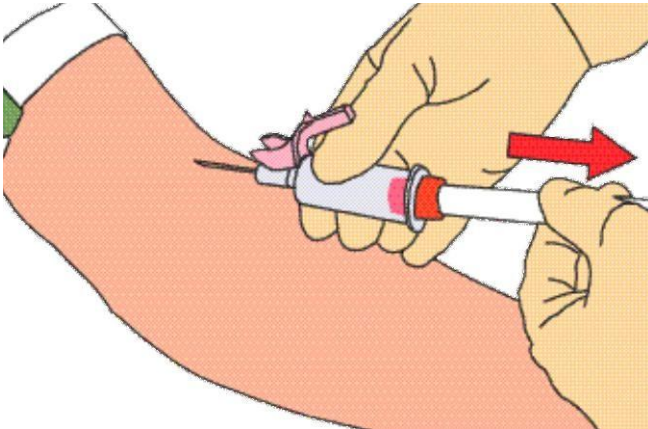
- Wrist

How to Take a Blood Sample

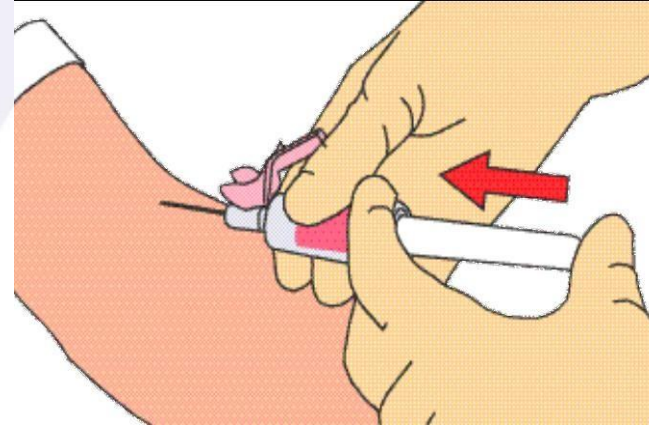
- Insert Needle



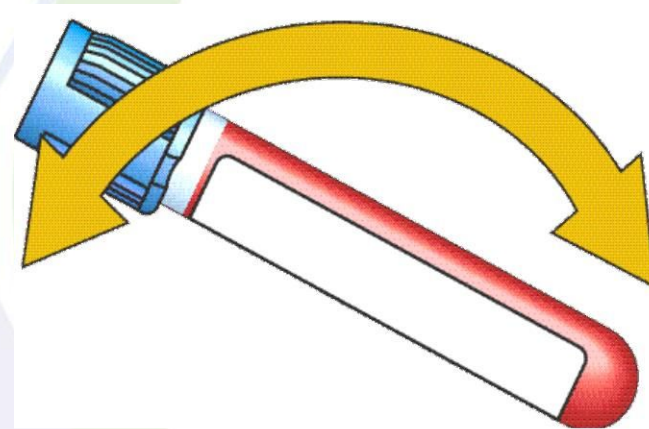
- Remove tube



- Introduce the tube

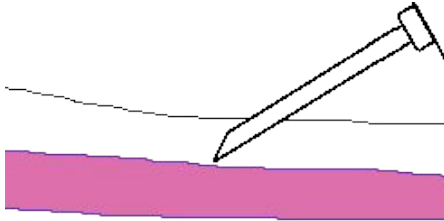


- 4-5 Gentle inversions

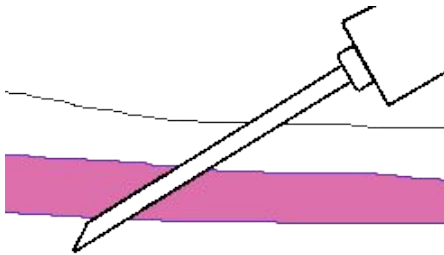


Trouble shooting:

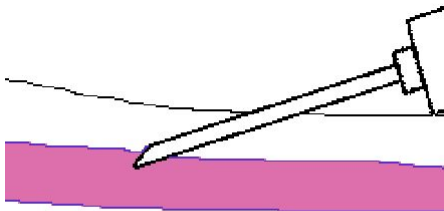
If an incomplete collection or no blood obtained



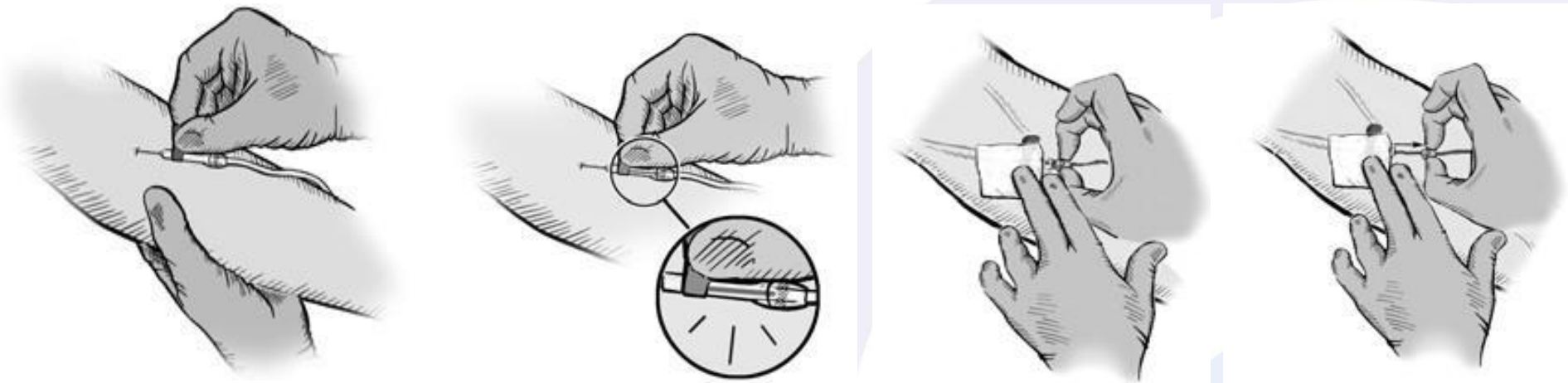
- No Flash back: Change position of needle
Move it forward - Might not be in deep enough



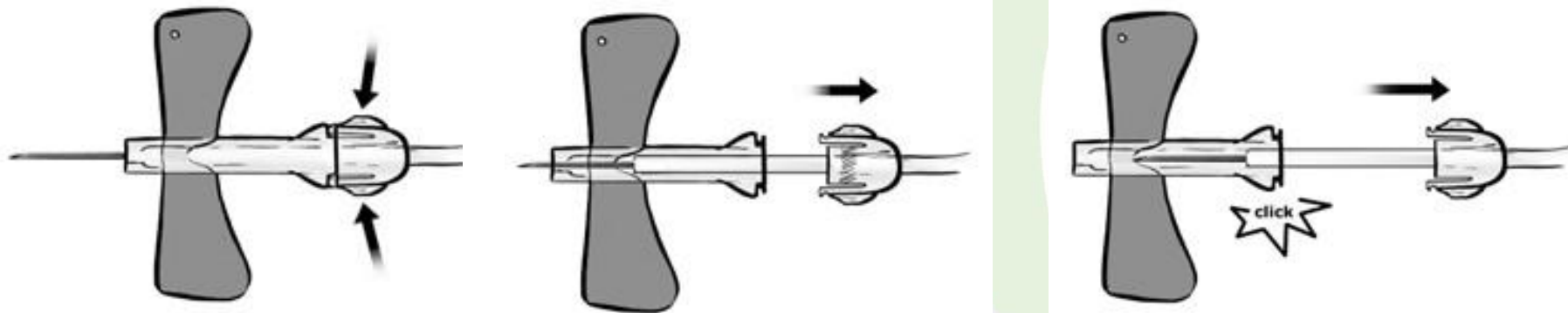
- Flash back but no fill: Change position of needle
Move it backward - Might be in too deep



- Flashback, poor fill: Adjust the angle - Bevel may be against vein wall



It is recommended to activate the safety mechanism when the blood collection has been completed and the VACUETTE® Safety Blood Collection Set is in the patient's vein.



Specimen Rejection

- i. Unlabeled specimen
2. Misabeled specimen
3. Inadequate Volume
4. Wrong Tube
5. Hemolysis
6. Clotted Anticoagulated Tube
7. Improper Transport
8. Missing Requisition Form
9. Contaminated Specimen

Common Errors

Identification of patient

Not labeling the tubes

Wrong name of the patient on tubes

Improper selection of tubes

Improper site cleansing

Improper specimen collection

Not properly mixing the tubes

Improper transportation of specimens

Problem Patients

1. **Fainting**
2. **Refusal**
3. **Nausea or Vomiting**
4. Convulsions
5. Excessive Bleeding
6. Cessations of Breathing
7. Unconsciousness
8. **Obesity**
9. **Tremors**
10. **Disruptive Behavior**
11. Deaf or Hearing Impaired
12. Special Precautions
13. Special Instructions at the Bedside
14. Blind or visually Impaired
15. **Hematoma**

Interpretation of Blood Results

- The medical team are responsible for interpreting blood results
- Where values are abnormal assess patient and prompt the Doctors to take appropriate clinical interventions
- Always ensure that the Doctor is informed of blood results, to action treatment as soon as possible

Questions?

